Exam A — 036026 Kinematics, Dynamics and Control of Robots

Spring Semester 2025

Exam information and instructions

- Duration: 14:00 - 17:00, June 23, 2025
- You are allowed to use only the lab computer at the workstation allocated to you. You are not allowed to use your own laptop or a tablet (like an iPad).
- The first 30 minutes of the exam is closed book, where you need to answer question 1 on Moodle, using the Safe Exam Browser. You are allowed to use a calculator to answer question 1. Question 1 consists of five short questions counting 3 marks each. You have only one opportunity to submit your answers; no repeated submissions are allowed. After 30 minutes (at 14:30) question 1 closes on Moodle and you will no longer have access to it. Make sure you submit your answers before it closes.
- After question 1 closes, the exam proceed to an open book part for the next 150 minutes, during which you are allowed to use paper notes and books. You are allowed to load data on the computer from a USB flash drive or external drive. You are also allowed to download files from your own space on Moodle. You may start with the open book part of the exam during the first 30 minutes of the exam, but you are not allowed to use your notes or books during the first 30 minutes.
- You are not allowed to use the internet except for uploading files or information to or from Moodle. You are not allowed to communicate with another person via email, WeChat or whatever other means possible. If you are caught using the internet — except Moodle — or communicating with a person other than the invigilators or the lecturer during the exam, this will be considered as academic dishonesty and disciplinary action against you will follow.
- You need to download various files that you need to use in answering questions 2, 3 and 4, from Moodle. All these files are in the zip file “Exam_A_files.zip”, which is available on Moodle under the “Exams” topic.
- The answers you present to the open book part of the exam should show a logical train of thought, starting from the theory that was covered in the reading assignments. If you use your computer to solve your problems, remember that the grader of your exam does not see your computer script and output. You must therefore supply enough written information in your exam script to allow the grader to assess your answer. If you solve equations symbolically or numerically in MATLAB or Python, these equations must be written in your exam script, with an appropriate and logical explanation of how they were derived.
- In final answers to questions, all fractions, products, sums, exponential expressions and natural constants like π should be evaluated to single real numbers, given accurately to four significant digits.
- For questions 2 and 3 you need to fill in variables or parameters that you have calculated in a feedback tool on Moodle. All the variables/parameters that you fill in in the feedback

tool, you must also clearly write in your exam script, so that it can be confirmed after the exam that the values entered on Moodle are actually the values that **you** have determined in the exam. If you fail to supply these parameters in your script, your answers on Moodle will be ignored and you will get zero for the corresponding part of the question. If you write the parameters in your script only, you will certainly loose marks.

- In all questions use $g = 9.81$ m/s^2 where required.
- The notation used in this exam is the same as what is used in the course text book by Franklin and in the course presentation, unless indicated otherwise.
- The paper consists of four questions. Full marks are 100.

Question 1 (15 marks)

Question 1 is asked as a quiz on Moodle, under the "Exam" topic. This question needs to be answered on Moodle, using the safe exam browser. This part of the exam is closed book and has a duration of 30 minutes.

Turn the page for question 2.

Revised Question 2 **(35 marks)**

The origin of the body frame B of a rigid body has coordinates $[\ 1.8 \quad -0.2 \quad 0.05\ ]^T$ in the inertial frame A. The directions of the axes of frame B with respect to the inertial frame is defined by the rotation matrix R_{ab}. The components of R_{ab} is given to you in double precision in the Excel spreadsheet file "q2dat.xlsx", in the "Exam_A_files.zip" file on Moodle, under the "Exam" topic, or as indicated below:

```
 9.1168461167710357e-01   3.2205442693477448e-01  -2.5517075640222342e-01
-2.2792115291927589e-01   9.1309336395051188e-01   3.3810125223294596e-01
 3.4188172937891381e-01  -2.5008289585905735e-01   9.0585618522789324e-01
```

If you choose to read the data from the file into MATLAB, you can use the command

```
Rab = readmatrix('q2dat.xlsx')
```

(a) The body is now displaced so that the origin of the body frame has coordinates $[\ 0 \quad -1.2 \quad -0.05\ ]^T$ in the inertial frame, while its x and y axes point toward the negative x and negative y directions of the inertial frame, respectively. To avoid confusion, we refer to the body frame after this displacement as frame C. Calculate the rigid transformation matrix $g_{bc} \in SE(3)$ associated with the motion described above. Explain in your script how you calculate your result, and report on the elements of g_{bc} both in your script and in the "Exam A Question 2 variables" feedback tool on Moodle. (On Moodle you need to report only on the 1st three rows of g_{bc}).

(b) It is valid to interpret the transformation g_{ab} as the displacement of frame B form an original position where its origin coincided with that of frame A and the axes of the body frame were aligned and in the same direction as those of frame A. Calculate the twist coordinates $\xi \in \mathbb{R}^6$ corresponding to the transformation g_{ab}, expressed in the inertial frame. Explain in your script how you calculate your result, and report on the elements of ξ both in your script and in the "Exam A Question 2 variables" feedback tool on Moodle.

(c) Calculate the twist coordinates $\xi^b \in \mathbb{R}^6$ corresponding to the transformation g_{ab}, expressed in the body frame B. Explain in your script how you calculate your result, and report on the the elements of ξ^b both in your script and in the "Exam A Question 2 variables" feedback tool on Moodle.

(d) Calculate the axis, the pitch and the magnitude of the screw associated with the twist ξ. The axis should be defined by a point $\in \mathbb{R}^3$ on this axis plus a unit vector $\in \mathbb{R}^3$ that indicates the direction of the axis. (Any point that lies on the axis may be specified.) The pitch and the magnitude are both scalar values. Explain in your script how you calculate your result, and report on these eight variables both in your script and in the "Exam A Question 2 variables" feedback tool on Moodle.

Turn the page for question 3.

Question 3 (35 marks)

Consider the 2-DOF manipulator shown in figure 3 in the reference configuration. In the reference configuration

- the centrelines of link 1 of length l_1 and link 2 of length l_2 are aligned (concentric) and parallel to the y axis of the spatial frame S,
- the tool frame T origin is a distance $l_1 + l_2$ away from the spatial frame z axis, while the respective axes of T are parallel to the corresponding axes of the spatial frame, and
- point p_b has coordinates $[\ 0 \quad 0 \quad l_0\ ]^T$ in the spatial frame.

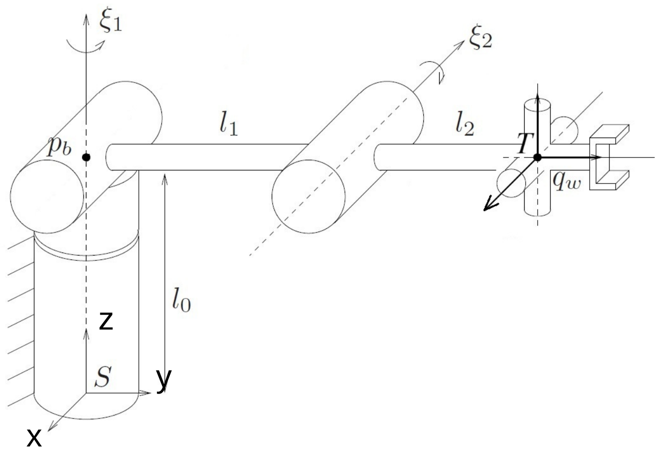

Figure 3

The manipulator is actuated and moves so that the configuration of the tool frame can be described by $g_{st} \in SE(3)$, given in components in the spatial frame, in double precision, in the Excel spreadsheet file “q3dat.xlsx”, in the “Exam_A_files.zip” file on Moodle, under the “Exam” topic, or as indicated below in single precision (which may work in this case, but it depends on your calculation):

$$
\begin{array}{rrrr}
0.84804810 & -0.28861472 & -0.44442769 & -0.56824277 \\
0.52991926 & 0.46188010 & 0.71123298 & 0.90937853 \\
0.00000000 & -0.83867057 & 0.54463904 & 0.080664716 \\
0.00000000 & 0.00000000 & 0.00000000 & 1.0000000
\end{array}
$$

If you prefer to read the data from the file into MATLAB (this is recommended), you can use the command

```
gst = readmatrix('q3dat.xlsx')
```

Calculate the joint angles θ_1 and θ_2 through which the revolute joints 1 and 2, respectively, rotated, in radians, to bring the tool frame to configuration g_{st} from the reference configuration. Explain in your script how you calculate your result, and report on these two angles both in your script and in the “Exam A Question 3 variables” feedback tool on Moodle.

Turn the page for question 4.

Question 4 (15 marks)

Consider the 2-DOF manipulator shown in figure 4 in the reference configuration. In the reference configuration

- the centrelines of link 1 of length l_1 and link 2 of length l_2 are aligned (concentric) and parallel to the y axis of the spatial frame S,
- the tool frame T origin is a distance $l_1 + l_2$ away from the spatial frame z axis, while the respective axes of T are parallel to the corresponding axes of the spatial frame, and
- point p_b has coordinates $[\ 0 \quad 0 \quad l_0\ ]^T$ in the spatial frame.

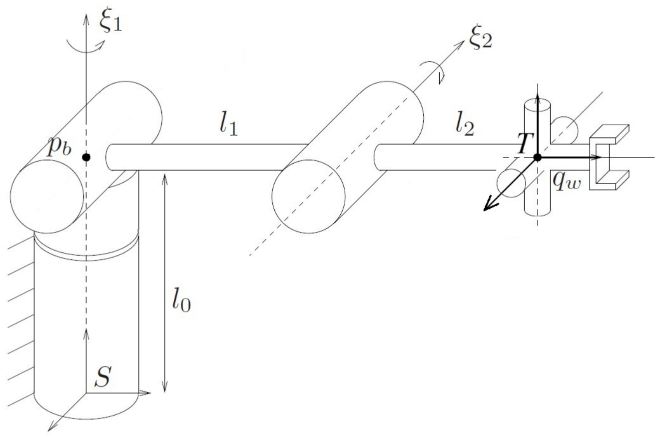

Figure 4

Calculate the dynamic response of the manipulator, for a 50 second period, after it is released from rest in the reference configuration, for three different applied joint torque cases; in all cases the torque is constant with time:

(a) $\tau = [\ 0 \quad 0\ ]^T$ Nm

(b) $\tau = [\ 0.04 \quad 0\ ]^T$ Nm

(c) $\tau = [\ 0.04$ -m2gr2 $]^T$ Nm

where m2gr2 is the torque of the weight of link 2 about ξ_2.

For each case you need to upload two graphs to the "Exam A Question 4" assignment on Moodle, firstly of $\theta_1(t)$ and secondly of $\theta_2(t)$, both on the vertical axis, in radians, against time in seconds on the horizontal axis. You also need to write a **short** interpretation of the graphs that you have uploaded to Moodle, in which you need to mention whether the results make sense, and if so, why. From this description it should also be clear that you yourself actually uploaded the graphs to Moodle. Note that you need to upload a total of six graphs to Moodle. Each graph should be clearly identified with the applied torque case used to generate them.

The following files are provided for your use in question 4 and are included in the "Exam_A_files.zip" file on Moodle, under the "Exam" topic:

- The file `MCandN_init.mlx`, which contains all the parameters of the manipulator arm and sets up everything that enables the calling of the function `MCN`. Executing this code in

MATLAB can also serve to initialize a SIMULINK program that would use the data. Note that the torque m2gr2 is actually defined in the file `MCandN_init.mlx`.

- The file `MCN.m`, which contains the function `MCN` that calculates the $M(\theta)$, $C(\theta, \dot{\theta})$ and $N(\theta, \dot{\theta})$ matrices that appear in the differential equations of motion of the manipulator, as derived in the course textbook by Murry, Li and Sastry.
- The SIMULINK template `ques4sim_templateR2023a.slx`, which merely contains the function `MCN`, ready for wiring up to a SIMULINK program.

You can calculate the dynamic response either by generating a SIMULINK program that solves the differential equations of motions, or by using an ordinary differential equation solver (like `ODE45`) directly in MATLAB.

If you find an aspect in the above files that are supplied to you, with which you do not agree, still answer the question with the files given as if they are correct.